

Philipp Becker

MACHINE LEARNING RESEARCHER

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Key Research Expertise

- Developing Generative AI and World Models for embodied reinforcement learning and efficient text-to-image synthesis.
- Driving RL-first foundation models for embodied agents, focusing on unsupervised learning and robust multimodal representations for robotics.
- Designing scalable, uncertainty-aware State-Space Models (SSMs) and advanced architectures for complex time-series and RL.
- Advancing LLM Alignment through a novel policy learning algorithm that significantly improves training stability and sample efficiency.
- Engineering Information-Theoretic RL methods to ensure stable, safe, and versatile control policies for real-world robotic deployment.

Experience

Postdoctoral Researcher (Research Scientist)

Paris, France

META SUPERINTELLIGENCE LABS - FUNDAMENTAL AI RESEARCH (FAIR)

10/2025 - 04/2027

- Conducting fundamental research towards an RL-first approach to Vision-Language-Action (VLA) models.
- Advancing representation and world modeling techniques to improve the efficiency and generalization of embodied agents.
- Accelerating the research lifecycle using agentic AI tools to rapidly prototype and scale training pipelines across multi-node GPU clusters.

Research Intern

Cambridge, United Kingdom

SAMSUNG R&D INSTITUTE UK, AI CENTER - CAMBRIDGE, GEN AI TEAM

11/2024 - 04/2025

- Researched linear attention for efficient, large multimodal diffusion transformers in text-to-image applications - results published at ICCV.
- Trained and evaluated generative AI models using multiple GPUs and nodes using distillation and fine tuning approaches.
- Working towards bringing state-of-the-art text-to-image models to mobile phones while contributing to the team's research codebase.

Doctoral Researcher

Karlsruhe & Tübingen, Germany

KARLSRUHE INSTITUTE OF TECHNOLOGY (KIT), AUTONOMOUS LEARNING ROBOTS LAB (SINCE 01/2020)

BOSCH CENTER FOR ARTIFICIAL INTELLIGENCE, ROBERT-BOSCH-GMBH (05/2019-12/2019)

05/2019 - 09/2025

- Published 4 first-author and 13 co-author publications at ICML, ICLR, NeurIPS, TMLR, RLC, RSS, CoRL, IROS, and ICRA.
- Supervised 25+ Bachelor and Master students, often with other labs or industrial partners, leading to 8 co-author publications.
- Ensured robust compute availability for efficient research in the lab by setting up and maintaining local GPU resources (SLURM-based cluster, desktop pool), supervising the development of experiment deployment tools, and coordinating with public providers of large-scale compute.
- Demonstrated organizational and problem-solving skills by supporting the establishment of two research labs and ensuring operational continuity during challenges such as the shift to remote lectures during COVID-19 and lab closures and relocations due to infrastructure issues.
- Organized, designed exams and exercises, and presented solutions for "Machine Learning - Foundations and Algorithms" (graduate level lecture, 150+ students) as the head teaching assistant for 3 years.
- Guest lectured on model-based reinforcement learning to 50+ students for "Reinforcement Learning" (graduate level lecture).

Research Scientist

Karlsruhe, Germany

FZI RESEARCH CENTER FOR INFORMATION TECHNOLOGY, (FZI FORSCHUNGSZENTRUM INFORMATIK)

04/2023 - 07/2024

- Worked towards establishing a new research group under the direction of Prof. Gerhard Neumann.
- Successfully evaluated structured reinforcement learning approaches for control applications of an industry customer.
- Contributed to securing funding for the EU Horizon project XSCAVE (€8M total, €950K to FZI from 2025 to 2028), aimed at advancing AI-driven planning and control for heavy machinery in industries like forestry and mining.
- Collaborated on research grant proposals with academic and industry partners, focusing on AI for production, energy, and heavy machinery.

Student Researcher

Darmstadt, Germany

TECHNICAL UNIVERSITY OF DARMSTADT, INTELLIGENT AUTONOMOUS SYSTEMS

04/2016 - 02/2019

- Developed a software stack for point cloud-based active segmentation with real and simulated robots in C++ using ROS and PCL.
- Collaborated on robotics projects, including a policy learning toolbox, feasibility studies on real robots, and setting up simulations.

Student Intern

Munich, Germany

'DEEP LEARNING AND ROBOTICS CHALLENGE' AT VOLKSWAGEN GROUP, DATA LAB IN COLLABORATION WITH NVIDIA

09/2017

- Developed a Lego robot with control and deep learning perception stacks for autonomous deployment on the Lego system and a NVIDIA Jetson.

Education

PhD (Dr. Ing.) in Computer Science, Machine Learning

Karlsruhe, Germany

KARLSRUHE INSTITUTE OF TECHNOLOGY, AUTONOMOUS LEARNING ROBOTS LAB

01/2020 - 10/2025

- Thesis: Probabilistic World and Behavior Models for Embodied Agents
- Advisor: Gerhard Neumann

Master of Science in Autonomous Systems (Computer Science), with Distinction

Darmstadt, Germany

TECHNICAL UNIVERSITY OF DARMSTADT

11/2016 - 04/2019

- Thesis: Expected Information Maximization: Using the I-Projection for Mixture Density Estimation.
- Honors: ECTS A Ranking, Deutschlandstipendium (Germany Scholarship), awarded for academic excellence (2016–2019).

Bachelor of Science in Computer Science

Darmstadt, Germany

TECHNICAL UNIVERSITY OF DARMSTADT

04/2014 - 10/2016

Summer Schools

- Nordic Probabilistic AI School, Trondheim, Norway, June 2023
- Machine Learning Summer School, Virtual (Tübingen, Germany), July 2020

First Author Publications

TROLL: Trust Regions Improve Reinforcement Learning for Large Language Models

ICLR (Oral)

P. BECKER*, N. FREYMUTH*, S. THILGES, F. OTTO, G. NEUMANN (* SHARED FIRST AUTHOR)

2026

EDiT: Efficient Diffusion Transformers with Linear Compressed Attention

ICCV

P. BECKER, A. MEHROTRA, R. CHAVHAN, M. CHADWICK, L. MORREALE, M. NOROOZI, A. G. C. P. RAMOS, S. BHATTACHARYA

2025

Combining Reconstruction and Contrastive Methods for Multimodal Representations in RL

RLC

P. BECKER, S. MOSSBURGER, F. OTTO, G. NEUMANN

2024

KalMamba: Towards Efficient Probabilistic State Space Models for RL under Uncertainty

Workshop, ICML & RL

P. BECKER, N. FREYMUTH, G. NEUMANN

2024

On Uncertainty in Deep State Space Models for Model-Based Reinforcement Learning

TMLR

P. BECKER, G. NEUMANN

2022

Expected Information Maximization: Using the I-Projection for Mixture Density Estimation

ICLR

P. BECKER, O. ARENZ, G. NEUMANN

2020

Recurrent Kalman Networks: Factorized Inference in High-Dimensional Deep Feature Spaces

ICML

P. BECKER, H. PANDYA, G. GEBHARDT, C. ZHAO, J. TAYLOR, G. NEUMANN

2019

Co-Author Publications (Selected, see)

Improving Long-Range Interactions in Graph Neural Simulators via Hamiltonian Dynamics

ICLR

T. HOANG, A. TRENTA, A. GRAVINA, N., P. BECKER, D. BACCIU, G. NEUMANN

2026

AMBER: Adaptive Mesh Generation by Iterative Mesh Resolution Prediction

NeurIPS

N. FREYMUTH, T. WÜRTH, N. SCHREIBER N., ..., P. BECKER, L. KÄRGER, G. NEUMANN

2025

Geometry-aware RL for Manipulation of Varying Shapes and Deformable Objects

ICLR (Oral)

T. HOANG, H. LE, P. BECKER, V. ANH NGO, G. NEUMANN

2025

Efficient Off-Policy Learning for High-Dimensional Action Spaces

ICLR

F. OTTO, P. BECKER, V. ANH NGO, G. NEUMANN

2025

Adaptive World Models: Learning Behaviors by Latent Imagination Under Non-Stationarity

Workshop, NeurIPS

E. GOSPODINOV, V. SHAJ, P. BECKER, S. GEYER, G. NEUMANN

2024

PointPatchRL - Masked Reconstruction Improves Reinforcement Learning on Point Clouds

CoRL

B. GYENES, N. FRANKE, P. BECKER, P., G. NEUMANN

2024

MuTT: A Multimodal Trajectory Transformer for Robot Skills

IROS

C. KIENLE, B. ALT, O. CELIK, P. BECKER, D. KATIC, R. JÄKEL, G. NEUMANN

2024

Iterative Sizing Field Prediction for Adaptive Mesh Generation From Expert Demonstrations

Workshop, ICML

N. FREYMUTH, P. DAHLINGER, T. WÜRTH, P. BECKER, A. TARANOVIC, O. GRÖNHEIM, L. KÄRGER, G. NEUMANN

2024

Beyond Deep Ensembles: A Large-Scale Evaluation of Bayesian Deep Learning under Distribution Shift

NeurIPS

F. SELIGMANN, P. BECKER, M. VOLPP, G. NEUMANN

2023

Information-Theoretic Trust Regions for Stochastic Gradient-Based Optimization

Workshop, NeurIPS

P. DAHLINGER, P. BECKER, M. HÜTTENRAUCH, G. NEUMANN,

2023

Accurate Bayesian Meta-Learning by Accurate Task Posterior Inference	ICLR
M. VOLPP, P. DAHLINGER, P. BECKER , C. DANIEL, G. NEUMANN	2023
Curriculum-Based Imitation of Versatile Skills	ICRA
M.X. LI, O. CELIK, P. BECKER , D. BLESSING, R. LIOUTIKOV, G. NEUMANN	2023
Inferring Versatile Behavior from Demonstrations by Matching Geometric Descriptors	CoRL
N. FREYMUTH, N. SCHREIBER, P. BECKER , A. TARANOVIC, G. NEUMANN	2022
End-to-End Learning of Hybrid Inverse Dynamics Models for Precise and Compliant Impedance Control	RSS
M. REUSS, N. VAN DUIJKEREN, R. KRUG, P. BECKER , V. SHAJ, G. NEUMANN	2022
Hidden Parameter Recurrent State Space Models For Changing Dynamics Scenarios	ICLR
V. SHAJ, D. BÜCHLER, R. SONKER, P. BECKER , G. NEUMANN	2022
Specializing Versatile Skill Libraries using Local Mixture of Experts	CoRL
O. CELIK, D. ZHOU, G. LI, P. BECKER , G. NEUMANN	2021
Differentiable Trust Region Layers for Deep Reinforcement Learning	ICLR
F. OTTO, P. BECKER , V. ANH NGO, H. ZIESCHE, G. NEUMANN	2021
Switching Recurrent Kalman Networks	Workshop, NeurIPS
G. NGUYEN-QUYNH, P. BECKER , C. QIU, M. RUDOLPH, G. NEUMANN	2021
Versatile Inverse Reinforcement Learning via Cumulative Rewards	Workshop, NeurIPS
N. FREYMUTH, P. BECKER , G. NEUMANN	2021
Action-Conditional Recurrent Kalman Networks For Forward and Inverse Dynamics Learning	CoRL
V. SHAJ, P. BECKER , D. BUCHLER, H. PANDYA, N. VAN DUIJKEREN, J. TAYLOR, M. HANHEIDE, G. NEUMANN	2020

Reviewing

Machine Learning	TMLR, NeurIPS, ICLR, ICML, CoRL, RLC
Robotics	RA-L, IROS, ICRA

Skills

Programming	Python (Expert), Deep CS Fundamentals (C/C++, Java background)
ML Frameworks and Scaling	PyTorch, Multi-GPU/Multi-node training, JAX, TensorFlow
AI-Accelerated Development	Agentic Workflows (Claude Code, Manus), LLM-Assisted Research Prototyping
Infrastructure and Robotics	Linux, Slurm, Docker, WandB, Git, MuJoCo, Real-World Robot Deployment (Franka, KUKA)